15

Substituting this into *(5)*, we have

$$\frac{2S}{t^2} = \frac{mg - \frac{T_f}{r}}{\frac{I}{r^2} + m} \quad \text{or} \quad \frac{2S}{t^2}\left(\frac{I}{r^2} + m\right) = mg - \frac{T_f}{r}$$

giving $mr\left(g - \frac{2S}{t^2}\right) = \frac{1}{t^2}\left(\frac{2S}{r}I\right) + T_f$ .....(7)

From equation (7) we see that a plot of $rm\left(g - \frac{2S}{t^2}\right)$ against $\frac{1}{t^2}$ gives a straight line slope of $\frac{2S}{r}I$ and an intercept T_f.

RESULTS: Tabulate your observations and determine the mean time taken in each case. Calculate the points to be plotted on a graph as above and also tabulate the values of I obtained directly from (7) with a knowledge of T_f. One sample calculation is sufficient.

PROCEDURE: Make sure the flywheel runs freely in its bearings and lubricate if necessary. Measure the diameter of the spindle and the cord to determine r. Check your value of r by a different method. With the cord long enough to allow the weight to touch the floor. Raise the weight to a known height and take the time for the weight to fall to the floor when released from rest. Check each reading two or three times. Repeat for different values of the falling weight. Set the flywheel turning and count the number of revolutions and the time taken for it to come to rest.

Repeat this several times to determine an independent estimate of T_f.

CONCLUSIONS: Compare the values of T_f obtained directly and from the graph. Would you expect any difference between these? Determine a mean value of I obtained directly from (7) and from the slope of the graph. State the limits of accuracy to be expected from the experiment. Compare the experimental with an estimate based on the dimensions and density of the material of the flywheel.

1. Does your estimate lie within the range of values suggested by the experiment?
2. What special precautions or difficulties should be noted?
3. How would your results be affected if the flywheel were unbalanced in its bearings?

Dr. J. Antonio
11/3/96

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SCHOOL OF ENGINEERING

MECHANICAL ENGINEERING DEPARTMENT
MECHANICS OF MACHINES LABORATORY INSTRUCTION SHEET

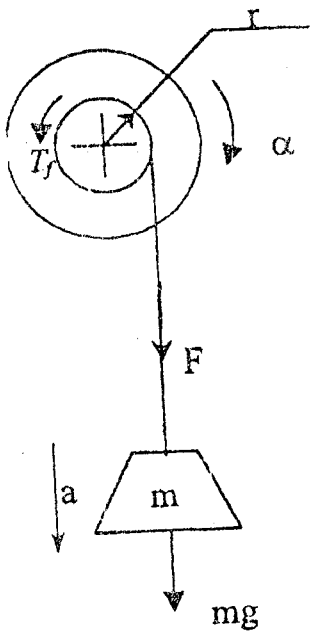
FLYWHEEL AND FALLING WEIGHT

OBJECT: To determine the moment of inertia of a flywheel by timing a falling weight.

APPARATUS: Flywheel mounted in horizontal low friction bearings, weights, string, rule, callipers and a stop-watch.

THEORY: For angular motion about a fixed axis,

Torque = Moment of inertia × Angular acceleration



For linear motion of a rigid body,

Force = mass × acceleration.

For the angular motion of the flywheel we have

$$Fr - T_f = I\alpha \ldots\ldots\ldots (1)$$

where r is the radius of the axle (with the thickness of the string taken into account), T_f, is the frictional torque and F is the tension in the string.

For the falling weight, we have;

$$Mg - F = ma \ldots\ldots\ldots (2)$$

where m is the mass of the falling weight and a is the acceleration. Noting that the angular acceleration $\alpha = \frac{a}{r}$ and eliminating F between equation (1) and (2) we have

$$mgr - T_f = I\frac{a}{r} + mar \quad \text{or} \quad \frac{mg - \frac{T_f}{r}}{\frac{I}{r^2} + m} = a \ldots\ldots (5)$$

Now, a is constant with time for any particular mass. If S is the distance descended from rest by the falling weight in time t, then the acceleration is given by,

$$a = \frac{2S}{t^2} \quad \text{(i.e., by applying the equation } S = ut + \frac{1}{2}at^2\text{)}$$

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CALIBRATION OF ORIFICES

THEORY

When liquid flows from an orifice under a head of h m, the calculated speed V_1 of the issuing jet is $V_t = 2gh\ (m/sec)$

If A m^2 is the area of the orifice, supposing that the cross sectional area of the jet is the same size, the calculated discharge. Q of liquid per second is $Q = A_1V_1\ m^3/sec.$

Actually, because of friction, the speed V of the jet is less than V_1

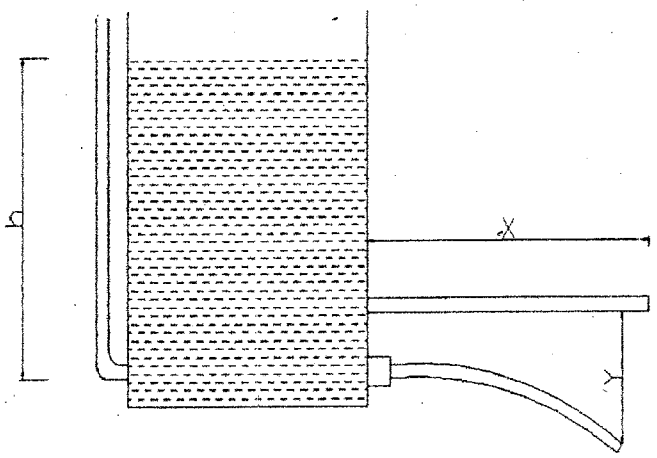
The ratio $\frac{V}{V_1}$ is called the coefficient of the orifice and is denoted by C_V. Similarly, the cross sectional area (A) of the jet is considerably less than the area of the orifice. The ratio $\frac{A}{A_1}$ is called the coefficient of contraction and is denoted by C_c. The actual discharge Q = AV, and calling the ratio $\frac{Q}{Q_1}$ the coefficient discharge C_d we have:

$$C_d = \frac{Q}{Q_1} = \frac{a_t V}{a_t V_t} = C_V C_C$$

AIM

The purpose of the experiment is to determine the value of these coefficients for a particular orifice and to find out whether the coefficient varies with change of head. The contraction coefficient need not be measured experimentally, but can be determined from the velocity and discharge coefficients by the equation given above.

SETUP



PROCEDURE

1. A number of orifice plates, having orifices of different shapes and sizes can be fitted into the side of a large tank.
2. The tank is supplied with water through an adjustable valve, and has internal baffle plates which smooth out turbulence of the incoming water. The side of the tank has a gauge showing height of water in the tank.
3. Open the supply valve and adjust until the water in the gauge stands at some definite height say 30m. The supply must be adjusted to keep this head constant.
4. The water runs into a calibrated tank and the discharge is measured by finding the amount of water collected in a certain time as measured by a stop-watch.
5. To find the speed of the jet, the shape of the path traced out by the jet should be measured. A horizontal scale is fixed above the jet. By measuring downwards form this, the vertical distance y between the scale and the upper surface of the jet can be found, at any horizontal distance x from the orifice.
6. Find y for values of x = 15cm, 30cm and 45cm etc. and measure also H_o corresponding to x = 0

Tabulate the results as below

Qty (l)	t (s)	$H_o(mm)$	$Qx10^{-4}(m^3/s)$	$H_o^{1/2}\ (m^{1/2})$
20				
20				
20				
20				
20				
20				
20				
20				
20				
20				

GRAPHS

Plot $Qx10^{-4}(m^3/s)$ against $H_o^{1/2}\ (m^{1/2})$ and determine the values of the coefficients C_d, C_V and C_C

QUESTIONS

1. What suggestions do you have for improving the apparatus?
2. If the graphs of Q against $\sqrt{H_o}$ does not appear to pass through the origin, what possible reasons are there for this? How will the value of C_d obtained from the graph be affected?
3. It is measured that the tank is sufficiently large for the velocity of flow towards the orifice to be neglected. Is this justified? If for example, the plan area of the tank was $4x10^{-2}m^2$. What is the mean velocity of approach when the discharge rate is $2x10^{-4}m^3/s$?
4. What is the velocity head corresponding to this velocity, and is it significant in relation to typical values of H_o used in the tests?

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CALIBRATION OF A VENTURI METER

INTRODUCTION

The Venturi Tube is a device used for measuring the rate of flow along a pipe. A fluid moving through it accelerates in the direction of the tapering contraction with an increase in the velocity in the throat. This is accompanied by a fall in pressure, the magnitude of which depends on the rate of flow. The flow rate may therefore be inferred from the difference in pressure as measured by piezometers placed upstream at the throat.

AIM

1. To obtain the calibration curve for the Venturi Meter.
2. To investigate the variation in pressure at the inlet and the throat at various rates of flow.
3. To present the results in a non-dimensional form so that they could be used to estimate the flow through any similar meter.

THEORY

(Refer to fig 7.1 and fig 7.2)

Consider the flow of an incompressible and inviscid fluid through the convergent-divergent venturi tube. Given that both the velocity and the piezometer head are constant over each of the sections considered, we may assume the flow to be one-dimensional so that the velocity and the piezometer head vary only in the direction of the tube length.

Treating the convergent-divergent pipe as a stream-tube and applying the Bernoulli's theorem to section 1, 2, 3..., n and have

$$\frac{U_1^2}{2g} + h_1 = \frac{U_2^2}{2g} + h_2 = \frac{U_n^2}{2g} + h_n \quad \ldots\ldots\ldots 1$$

The continuity equation is given by

$$U_1A_1 = U_2A_2 = U_nA_n = Q \ldots\ldots\ldots\ldots 2$$

Substituting equation 1 for U_1 into equation 2 gives

$$\frac{U_2^2}{2g}\left(\frac{A_2}{A_1}\right)^2 + h_1 = \frac{U_2^2}{2g} + h_2 \ldots\ldots\ldots\ldots 3$$

$$U_2 = \sqrt{\frac{2g(h_1 - h_2)}{1 - \left(\frac{A_2}{A_1}\right)^2}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots 4$$

The rate of flow $Q = A_2 \sqrt{\frac{2g(h_1 - h_2)}{1 - \left(\frac{A_2}{A_1}\right)^2}}$ (for ideal discharge) 5

The actual discharge is given by

$$Q = CA_2 \sqrt{\frac{2g(h_1 - h_2)}{1 - \left(\frac{A_2}{A_1}\right)^2}} \quad \ldots\ldots\ldots\ldots\ldots 6$$

Where C is the discharge coefficient

The velocity head, $U_2^2/2g$ at the throat can be conveniently used to express a dimensionless way of expressing the distribution of the piezometric head along the length of the Venturi meter. Accordingly, the piezometric Head Coefficient

$$C_p h = \frac{h_n - h_1}{U_2^2/2g} \qquad (n = 2,3,4 \ldots.) \ldots\ldots\ldots\ldots\ldots 7$$

The ideal distribution of $C_p h$ along a Venturi meter (in terms of its geometry) is given by

$$C_p h = \left(\frac{A_2}{A_1}\right)^2 - \left(\frac{A_2}{A_n}\right)^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots 8$$

$$C_p h = \left(\frac{D_2}{D_1}\right)^4 - \left(\frac{D_2}{D_n}\right)^4 \quad \ldots\ldots\ldots\ldots\ldots\ldots 9$$

PROCEDURE

1. Level the apparatus by opening both the bench supply valve and the control valve downstream of the meter to allow water to flow and clear air pockets from the supply hose.
2. The control valve is gradually closed causing the water to rise up in the tubes of the manometer thereby compressing the air contained in the manifold.
3. When the water level has risen to a convenient height, the bench valve is also closed gradually so that as both valves are finally shut off, the meter is left containing static water at moderate pressure.
4. Operate the adjust screws to give identical readings for all the tubes across the whole width of the manometer board. To establish the meter coefficient measurements, a set of differential heads $(h_1 - h_2)$ and flow rates Q should be made. The first reading should be taken with the maximum possible valve if $(h_1 - h_2)$ i.e. with h_1 close to the top of the scale and h_2 near to the bottom. This is obtained by gradually opening both the bench valve and the control valve in turn.
5. Successive opening of either valve will increase both the flow and the difference between h_1 and h_2.
6. The flow rate is found by timing the collection of a known amount of water in the weighing tank, with the meantime valves of h_1 and h_2 being read from the manometer.
7. Similar readings are then taken over a series of reducing valves of $(h_1 - h_2)$ roughly equally spread over the available range from 250 mm to 0. About 10 readings should suffice.

The report for the experiment should include;

1. Objective of the experiment
2. Arrangement and description of apparatus
3. Description of method
4. Table of observed and derived results
5. Graph of $(h_1 - h_1)^{1/2}$ versus Q (measured flow rate)
6. Graph of Q versus differential head (i.e. calibration curve)
7. Graph of C versus Q
8. Variation of measured and ideal values of $C_p h$ along the Venturi meter i.e. Graph $C_p h$ (measured and ideal) versus X_n (mm) where X_n is distance of locations of piezometer tubes A,B,C,D,E,F,G,H,J,K,L with A as the datum.
9. Discussion of results and conclusion including the use of the calibration curve. If the Venturi meter axis were vertical, would you make any correction to the piezometer?

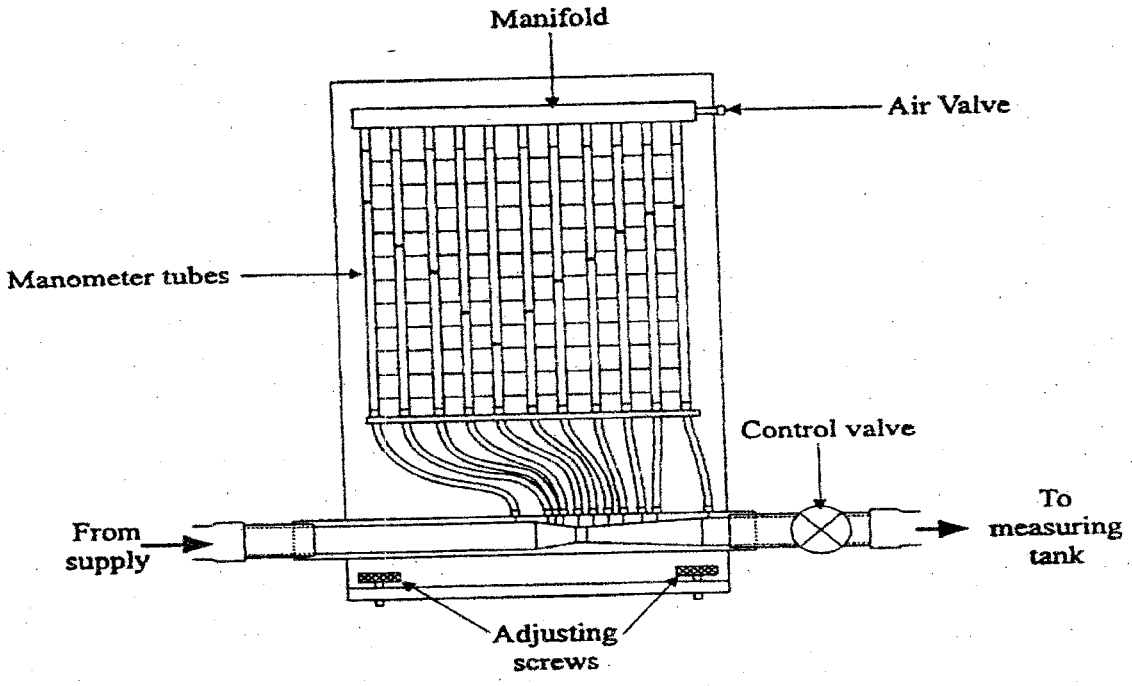


Figure 7.1 Arrangement of the Venturi meter apparatus

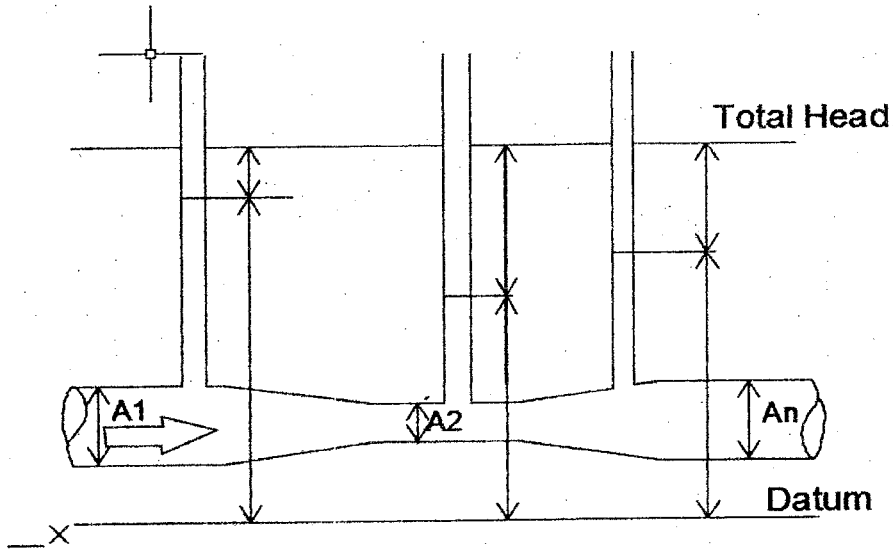


Figure 7.2 Arrangement of ideal Venturi meter

HYDRAULICS LABORATORY

REYNOLD'S NUMBER EXPERIMENT

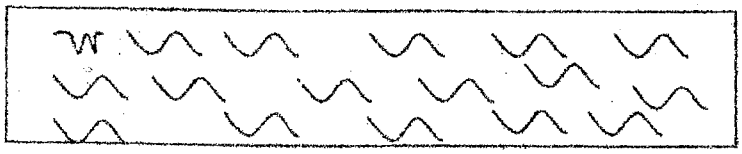
Objective:

1. To demonstrate the different flow,
   a. Laminar flow
   b. Transition
   c. Turbulent flow
2. To obtain Reynolds numbers corresponding to these flow regimes

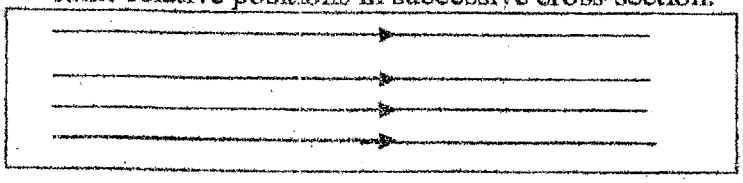
Introduction

Two types of flow occur. These are:

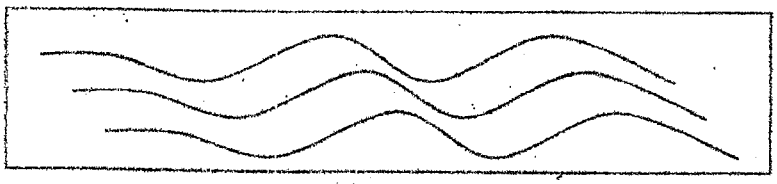
1. Turbulent flow: An unsteady flow, chaotic flow of particles with strong transverse mixing within the fluid. Thus fluid particles move in a disorderly manner, occupying different relative positions in successive cross section.



2. Laminar flow: the particles of the fluid move in an orderly manner and retain the same relative positions in successive cross-section.



There is also a type of flow where the fluid flow observed fall between laminar and turbulent.

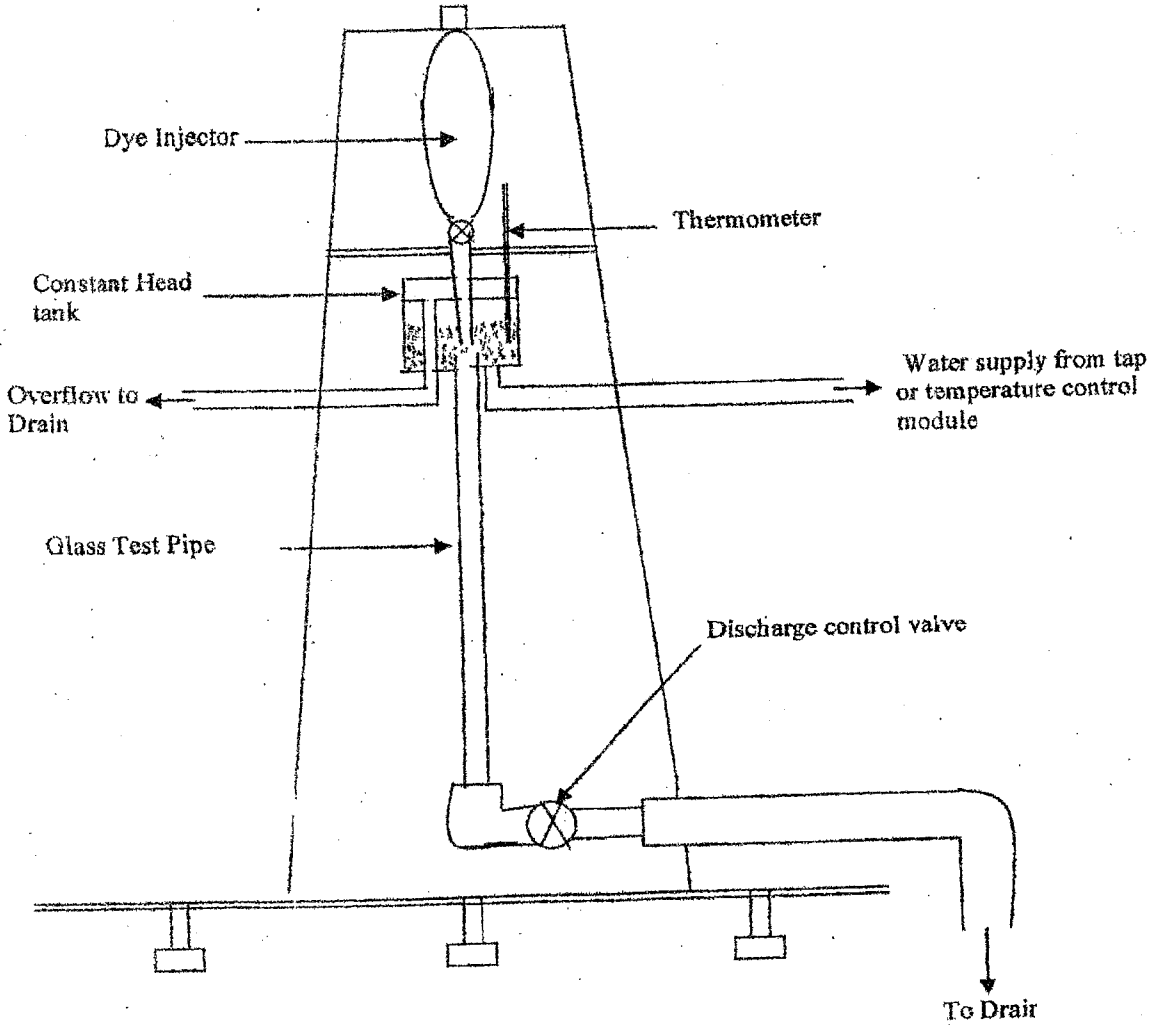


Osborne Reynolds found that the type of flow is determined by the velocity, density and viscosity of the fluid and the size of the conduct and depends on the value of $\frac{vd}{\vartheta}$.

v = velocity
d= typical dimension which for a pipe is the diameter
ϑ= viscosity of the fluid

For flow in pipes if the Reynolds number (Re) is less than 2100 (approx) flow is always laminar, if it is about 2100, there is a transitional region and for higher values of Re, flow will always be turbulent.

Experimental set-up



Procedure

1. The rate at which water flows through a tube is controlled by a tap B and its path is traced with the introduction of a dye from the U shaped reservoir.
2. Tap A is opened slightly and Tap B adjusted until the required flow is observed.
3. The level of water in the tank (rectangular tank) is kept constant by opening a tap into which flow at almost the same rate as water flow out.
4. The volume of water at any time (t) is taken by means of stop clock for any particular flow. The flow rate is increased by opening the tap B. Laminar flow changes to transition and further increase to turbulent, taking the same observation.

Theory
Derive the Reynolds's number $Re = \frac{vd}{\upsilon}$

Tabulate results as below:

Flow Type	Volume V (m^3)	Time t (s)	Discharge Q=V/T	Velocity V=Q/A	Re

Diameter of Tube (d)
Area of tube (A) = $\frac{\pi d2}{4}$
Temperature of water T °C =
Kinematic viscosity of water at T =

CONCLUSION
In conclusion, answer the following questions:
(1) The critical Re is 2100 and the velocity of flow corresponding to this critical Re is termed the critical velocity. Would you think that all fluids have the same critical velocity in a pipe of a given diameter? Give reasons.
(2) What is the use of Re in practice?
(3) Is flow rate affected by the value of Re?
(4) Does lamina flow occur in practice?
(5) The flow in river is laminae? True or False?
(6) Draw graph of Re verses discharge Q and V and discuss the results.
(7) Discussion should include how you tried to eliminate errors.

1.3 CALIBRATION OF AN INDICATOR SPRING AND PRESSURE GAUGE

OBJECTIVE

The object of the experiment is to determine the errors in the indicated pressures on a Bourdon type pressure gauge and to plot curves showing the corrections to be applied and to determine the spring constant for an indicator spring.

PROCEDURE

Connect the pressure to the test and fill the apparatus with oil. Take the gauge reading at zero pressure, put weights on the vertical piston to give a true pressure of 69 kNm^{-2} (101 $lbin^{-2}$) in the system and screw in the plunger until the vertical piston and weights rise slightly. Rotate the weights to minimize friction and take the gauge readings. Repeat in steps of 69 kNm^{-2} (101 $lbin^{-2}$) up to the maximum pressure for the pressure gauge. If the plunger becomes screwed right and during the test put more oil into the apparatus.

Repeat the test with the indicator in place of the pressure gauge. Obtain the pressure readings from the indicator by marking a card on the drum with the pencil for each pressure. **CALCULATE THE SPRING CONSTANT** in ($lbin^{-2}$) and compare it with the value printed on the spring 2.72×10^4 kNm^{-2} (100 $lbin^{-2}$).

For the pressure gauge, calculate the errors and plot;

1. "Straight Line" graph of indicated pressure against correction to be applied to obtain the true pressure.

2. "Straight Line" graph of indicated pressure against plotted by straight lines.

Comment on the suitability of the graphs for applying corrections in practice.
For the indicator, plot a curve of True pressure against Distance of point above the zero position.
This curve should be a straight line. The slope is the spring constant.

INDICATOR AND PRESSURE GAUGE LOG SHEET

TRUE PRESSURE	GAUGE READING	CORRECTIONS TO BE APPLIED	INDICATOR DISTANCE
10	12		
20	21		
30	31		
40	41		
50	51		
60	61		
70	71		
80	81		
90	91		
100	101		
110	111		
120	121		
130	131		
140	141		
150	152.5		
160	165		
170	171		
180	181		
190	191		
200	201		
210	201		
220	221		
230	231		
240	241		
250	251		

NB: Since the pressure is measured using the pressure gauge and is in lbs/in^2 they should be multiplied by the factor 6.895 to measure them in kN/m^2

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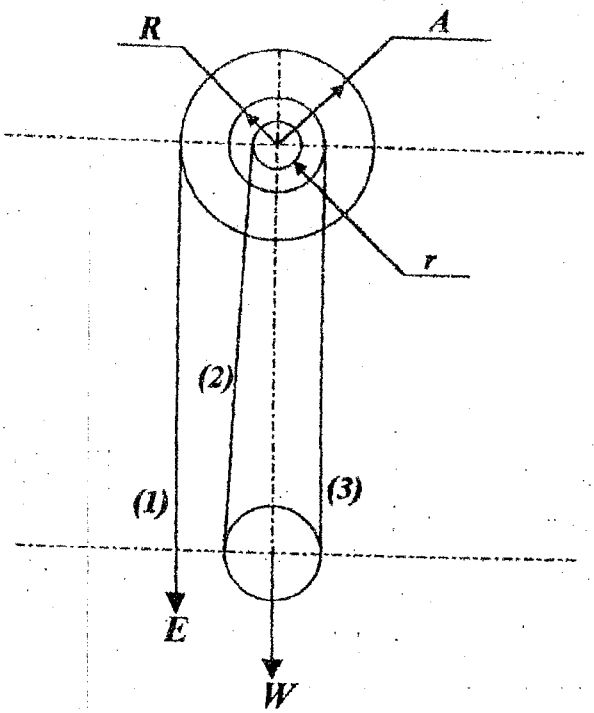
MECHANICAL ENGINEERING DEPARTMENT
MECHANICS OF MACHINES LABORATORY INSTRUCTION SHEET

THE DIFFERENTIAL PULLEY

OBJECT: To determine the mechanical advantage and velocity ratio for a differential pulley and to deduce the law of the machine.

APPARATUS: This shown in the diagram.

THEORY: This diagram shows a wheel of effective radius A, to the rim of which an *effort* is applied The wheel is keyed to the shaft having two different radii R and r.



A cord is fastened to the shaft and is so arranged that it winds unto one section of the shaft while winding off another. Suppose the wheel and the shaft turn anti-clockwise through an angle θ. The cord *(1)* descends $A\theta$, cord *(2)* descends $r\theta$, cord *(3)* rises $R\theta$. The resultant motion of the load W is the mean of the motion of cords *(2)* and *(3)* and hence it rises $(R-r)\frac{\theta}{2}$. So we can determine the velocity ratio as: $\frac{2A}{(R-r)}$.

If there were no friction or idle loads (such as the weight of the pulley or cord) then;

The *mechanical advantage* $\frac{W}{E}$ would be equal to the *velocity ratio* by the principle of conservtion of energy. In practice energy is wasted and so, for a rising *load*, W, $\frac{W}{E}$ is less than $\frac{2A}{(R-r)}$.

The ratio $\frac{Mechanical..Advantage}{Velocity..Ratio}$ is equal to the ratio $\frac{Work..done..on..load}{Work..done..by..effort}$, and is the *mechanical efficiency* of the machine.In this case this will be given by $\frac{W(R-r)}{2AE}$

PROCEDURE: Hang suitable series of *loads* on the hook attached to the centre of the pulley and determine the *effort* E to sustain steady motion, for each case. From the dimensions of the various parts measure the *velocity ratio* and compare this with the figure obtained for direct measurement of the movement of E and W. Tabulate *load*, *effort* and *mechanical efficiency*. Plot curves of *load* and *mechanical efficiency* against *effort*. Determine the "law of the machine" in the form $E=aW+b$ and also determine an equation of suitable form relating *mechanical efficiency* and *effort*.

CONCLUSION: State the law of the machine and the *maximum efficiency* obtained. Comment on your results, particularly any points which do not fall on the smooth curves. Do you consider that the law you have determined will be applicable to all mechanisms and if not to what type of mechanisms can they be applied? Under what circumstances will the *load* be capable of raising the *effort* E (i.e., the mechanism operate in reverse)? Does this illustrate any general principles?

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